

Common Low Back Injuries and Conditions

Nonspecific Low Back Pain

Injury overview	This general term describes the collection of soft-tissue musculoskeletal injuries in sports, which typically involve the paraspinal muscles. Poor posture, inefficient movement patterns, and faulty training techniques elevate the risk of soft-tissue injuries to the lower back.
Surgical and rehabilitative management	Surgery is rarely required. Rehabilitation begins to control pain, inflammation, and loss of daily function. Then rehabilitation progresses to address flexibility deficits and improve posture/core stability, balance control, and functional strength.
Return-to-play exercise and conditioning considerations	Pain-free ROM of spinal flexion, extension, lateral flexion (to each side), and lumbar rotation (to each side). Establish comprehensive baseline ROM data to guide continued training and serve as a comparator in the event of reinjury.
Return-to-play criteria	Multiplanar and multi-segmental stability are primary focus areas.
Estimated timeline	Variable. One day to several weeks, depending on the degree of damage and reliance on low back area function during sport-specific tasks.

Disc Herniation

Injury overview	Intervertebral discs are positioned between vertebral segments and function to maintain consistent joint spacing and provide cushioning. Disc herniations (bulging) predominantly occur in the posterior aspect of the lumbar spine at L4–L5 and L5–S1. These segments have the greatest stress on the lumbar spine during forward bending and rotation activities. During the herniation, disc particulates enter the spinal column, creating mechanical pain and chemical irritation, elevating the pain and discomfort levels an athlete encounters.
Surgical and rehabilitative management	Conservative rehabilitation is typically initiated first, focusing on inflammation and pain control, then gradually increasing therapeutic exercises. Should this approach fail, surgical repair is recommended.
Return-to-play exercise and conditioning considerations	Base training programs on the core stability and neuromuscular training continuum to account for the athlete's ability to dynamically stabilize multiple joint segments at the right time, with the right amount of force.

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Return-to-play criteria	Pain-free ROM (flexion, extension, lateral flexion, and rotation). Completion of a sport-specific performance test battery to include the upper-extremity closed kinetic chain stability test. Goniometric values should be within 80% of normal for all hip measures and ankle dorsiflexion.
Estimated timeline	2 weeks to 2 months.
Spondylosis and Spondylolisthesis	
Injury overview	Commonly appearing in the lumbar spine (L5–S1), spondylolysis is a fracture defect in the vertebral arch. When the vertebral body below the defect slips anteriorly, the condition is known as spondylolisthesis. Either condition can contribute to low back pain and related movement dysfunction.
Surgical and rehabilitative management	Surgical repair is necessary for persistent symptoms or spine instability. Following surgery, the use of back braces is common to immobilize the area and allow the best opportunity for the fusion to heal.
Return-to-play exercise and conditioning considerations	Develop a comprehensive spinal stability program; however, avoid hyperextension-type exercises.
Return-to-play criteria	Normal, pain-free ROM and completion of a sport-specific performance test battery without complications.
Estimated timeline	2 weeks to 2 months (if a surgical repair was completed, this would extend an additional 6 months).